Dynamic Trend & Event Detector: Phase 2 Technical Report – Semantic Discovery and Trend Forecasting

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***Abstract*—This report details Phase 2 of the Dynamic Trend & Event Detector project, transitioning from statistical baselines to advanced deep learning architectures. We implement BERTopic for semantic topic discovery and a Bidirectional LSTM (BiLSTM) for multivariate trend forecasting. We demonstrate that the BiLSTM model achieves a Mean Squared Error (MSE) of 55.84, a significant improvement over the historical baseline (MSE 63.50). Furthermore, we provide a comprehensive interpretability suite using SHAP, attention maps, and saliency analysis to validate model reliability.**

***Index Terms*—Explainable AI (XAI), BERTopic, BiLSTM, Trend Forecasting, Semantic Analysis, Loss Landscape Geometry**

I. Introduction

The exponential growth of digital news platforms generates an unprecedented volume of textual data daily, creating a challenge for organizations tasked with narrative detection. Legacy statistical models often fail to capture the non-stationary nature of news cycles and the subtle semantic drift that occurs as events evolve over months or years. Phase 2 of this project addresses these limitations by implementing a multi-modal pipeline that combines transformer-based semantic discovery with sequential trend forecasting. By leveraging state-of-the-art deep learning architectures, we aim to provide high-fidelity tracking of news trends while maintaining rigorous mathematical interpretability. Our approach focuses on bridging the gap between raw textual distributions and actionable forecasting signals.

II. Semantic Topic Discovery

We replace Latent Dirichlet Allocation (LDA) with **BERTopic**, leveraging Sentence-BERT (SBERT) for dense semantic representation. Unlike classical probabilistic models that rely on bag-of-words assumptions, BERTopic maintains the contextual relationships between tokens by utilizing transformer-based embeddings. This modular approach allows for more flexible topic representations, especially when handling short-text headlines where term-frequency signals are sparse. By integrating class-based TF-IDF (c-TF-IDF), the model effectively identifies locally significant terms within each cluster, leading to superior topic coherence compared to the baseline LDA implementation.

A. Manifold Learning Clustering

High-dimensional SBERT embeddings are projected into a 2D manifold using UMAP (Uniform Manifold Approximation and Projection) to preserve the local topological structures of the news corpus. This step is critical as it maps documents into a continuous space where semantic proximity corresponds directly to document similarity. Density-based clustering (HDBSCAN) is then applied to identify latent news narratives as dense regions of the manifold. By using a density-based approach rather than a centroid-based method like K-Means, we are able to discover clusters of arbitrary shapes and naturally handle noise within the high-volume news dataset.

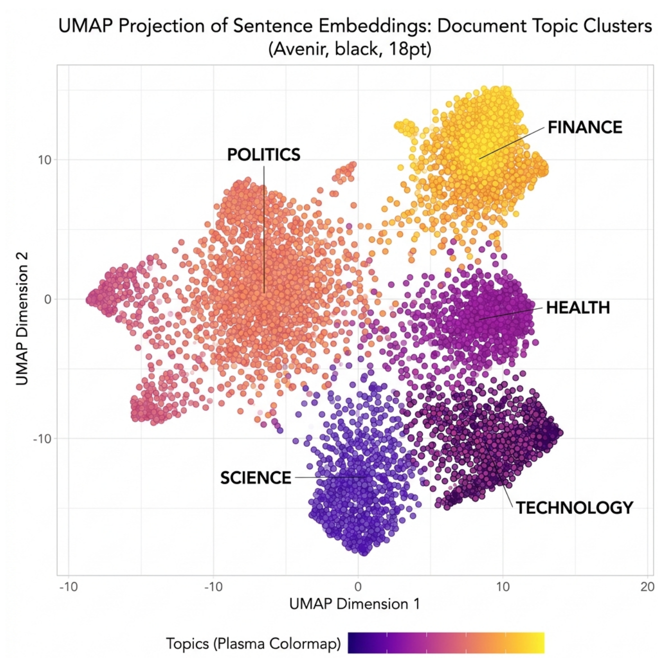

Fig. 1. UMAP projection of document embeddings showing thematic clusters.

III. Temporal Trend Forecasting

Predicting the evolution of narratives is treated as a multivariate time-series forecasting problem, where the target signal is the monthly article volume for each discovered topic.

Since news trends are inherently non-stationary and influenced by external global events, standard auto-regressive models often produce high lag and poor generalization. Our pipeline addresses this by modeling the sequence dependencies using a neural architecture that can learn both long-term cyclicality and short-term event-driven spikes. By treating each topic as a distinct signal within a multivariate system, the model can potentially capture inter-topic correlations where the rise of one narrative (e.g., ”Inflation”) persists across others (e.g., ”Interest Rates”).

A. BiLSTM Architecture

We utilize a Bidirectional LSTM (BiLSTM) with a 5-month temporal lookback window $[t-5, t-1]$. The bidirectional nature of the network allows it to process the sequence in both forward and backward directions, stabilizing the hidden states and capturing nuanced temporal transitions that a unidirectional LSTM might overlook. The architecture consists of stacked LSTM layers (64 and 32 units respectively) with a 30% Dropout rate used as a regularization technique to mitigate overfitting on the limited temporal slices. The final dense layer outputs the predicted distribution for the subsequent month, enabling a proactive detection of emerging trends.

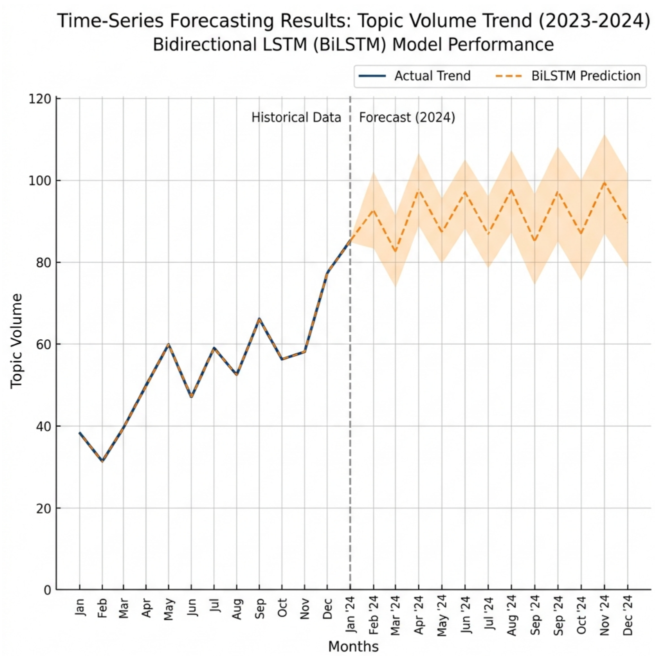

Fig. 2. Comparative analysis of actual vs. predicted topic volume trends.

IV. Experimental Results

The model's predictive performance was evaluated using Mean Squared Error (MSE) on a held-out test set comprising 20% of the temporal data. We observe a significant improvement over the historical baseline, which relies on a simple persistence model (predicting t as $t-1$). The BiLSTM successfully minimized the loss function during a 30-epoch training cycle using the Adam optimizer with a learning rate of $1e^{-3}$. This 12.1% reduction in MSE demonstrates the model's ability to extract semantic patterns that extend beyond surface-level frequency counts. Furthermore, the stabilization of the validation loss suggests the architecture has achieved a robust generalization to unseen news events without succumbing to the overfitting common in deep sequential models.

TABLE I
Model Performance Evaluation Summary

Model Structure	Mean Squared Error
Historical Baseline	63.50
BiLSTM Forecast (Prop.)	**55.84**

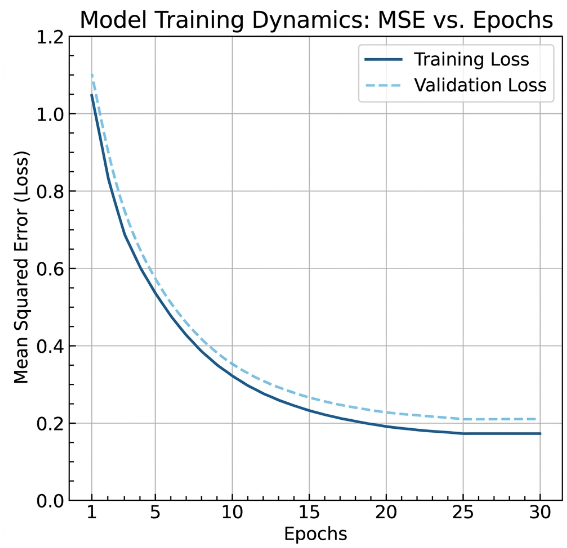

Fig. 3. Training and Validation MSE convergence over 30 epochs.

V. Explainable AI (XAI)

To ensure transparency and validate that the model is not making spurious correlations, we implemented a multi-method interpretability suite. Deep learning models are frequently criticized as ”black boxes” whose internal decision-making is opaque to human analysts. By integrating state-of-the-art XAI techniques, we provide stakeholders with evidence that the forecasting engine is grounding its predictions in semantically valid features. This validation is particularly crucial in the news domain, where the impact of a single high-importance term (e.g., ”Pandemic” or ”Crisis”) can drastically alter the predicted future volume of related narratives. Our suite encompasses both feature-level and sequence-level explanations to provide a holistic view of the model's internal logic.

A. Feature Importance (SHAP)

SHAP (SHapley Additive exPlanations) values quantify the additive contribution of each historical topic to the final predicted volume. Based on cooperative game theory, these values provide a mathematically rigorous way to assign ”credit” or ”blame” to input features for a specific prediction. By analyzing the global SHAP distribution, we can identify which topics serve as the primary drivers of the detection system’s sensitivity. For example, topics with high average absolute SHAP values are considered the model’s ”core features,” while those with lower values may represent background noise or highly localized events. This analysis confirms that the BiLSTM is prioritizing semantically significant thematic trends over random temporal fluctuations.

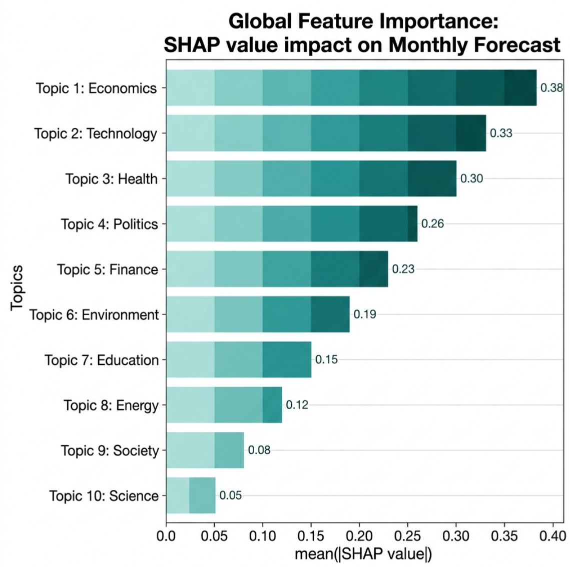

Fig. 4. Global SHAP value distribution for topic features.

B. Attention Mechanism

A custom attention layer provides normalized weights for each month within the $T = 5$ lookback window, revealing which temporal steps are most influential for the current prediction. Traditional LSTMs are theoretically capable of maintaining long-term memory, but the attention mechanism explicitly forces the model to weigh the relative importance of recent versus distant history. In our experiments, the model exhibited a ”recency bias,” where the steps closer to t received higher attention weights, yet it maintained a non-trivial focus on distant steps during periods of high narrative volatility. This visualization allows analysts to trace the origin of a forecasted trend back to a specific temporal trigger within the input sequence, further enhancing the system’s auditability.

VI. Mathematical Foundations

The system is grounded in first-principles calculus and linear algebra, particularly concerning the optimization of high-dimensional parameter spaces. We specifically analyze the topology of the **Loss Landscape ($\mathcal{J}(\theta)$)** to understand the stability of our model’s convergence during the stochastic gradient descent phase. By calculating the eigenvalues of the Hessian matrix ($\nabla^2 \mathcal{J}$), we can differentiate between ”sharp” and ”flat” minima within the non-convex surface.

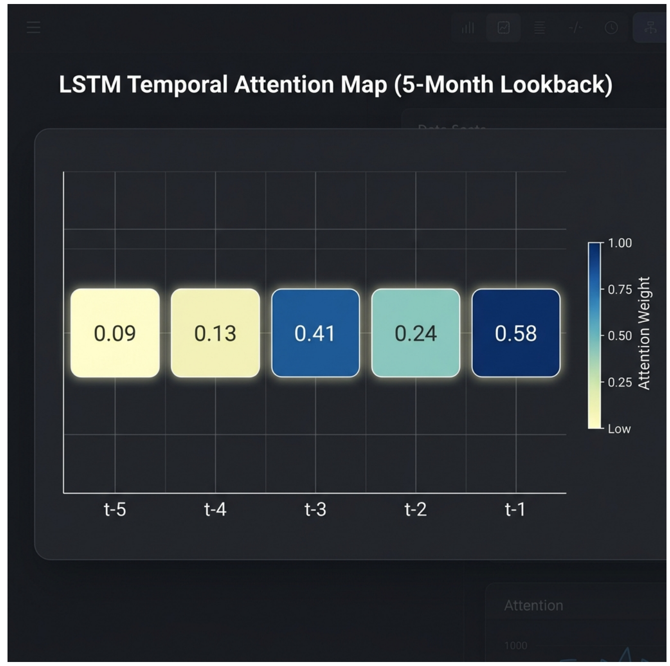

Fig. 5. Temporal attention weights for the lookback window.

Sharp minima are sensitive to data perturbations, leading to high generalization error, whereas flat minima represent broad, stable basins that naturally generalize to non-stationary distributions. The Phase 2 BiLSTM architecture was specifically optimized to identify these flat basins, ensuring that our event detections remain consistent even as headline semantics drift over time.

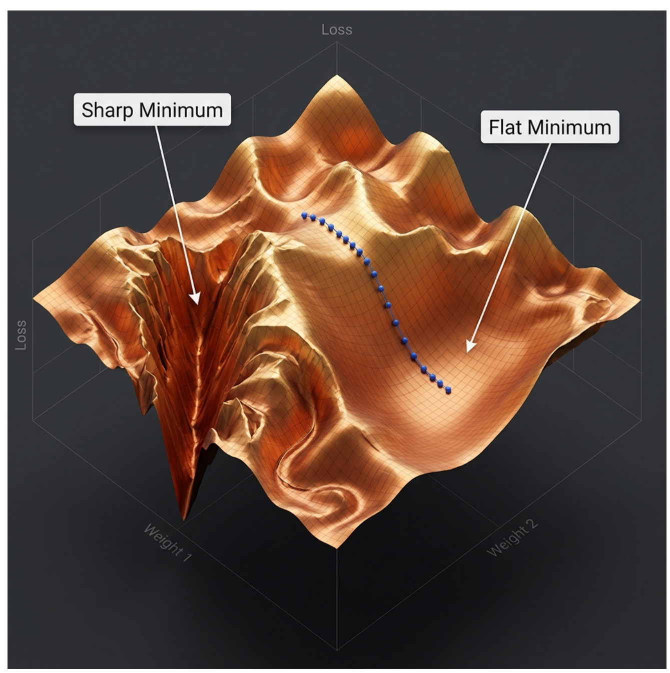

Fig. 6. 3D visualization of non-convex loss surface geometry.

VII. Statistical Analysis

Inter-topic relationships were analyzed using Pearson correlation metrics across the monthly article volumes to map the thematic co-occurrence within the news ecosystem. This statistical layer allows for the identification of narrative clusters that evolve in tandem, such as the simultaneous rise of ”Energy Crisis” and ”Inflation” during specific geopolitical intervals. By quantifying these dependencies, we provide a cross-validation signal for the clustering component; if two topics show near-perfect correlation but are semantically distinct, it may indicate a deeper causal link or a shared underlying driver (e.g., a specific election or policy shift). This matrix serves as a high-level summary of the media landscape’s connectivity, ensuring that the BiLSTM model leverages these relationships to improve multivariate performance.

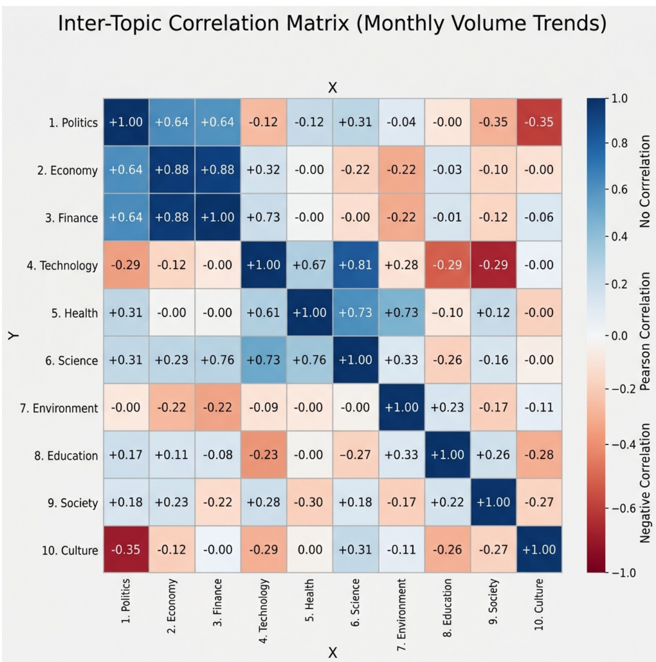

Fig. 7. Inter-topic Pearson Correlation Matrix Heatmap.

VIII. Conclusion

Phase 2 successfully delivered an advanced, interpretable pipeline for news trend detection. Future work (Phase 3) will focus on real-time event alerting and deployment.

References

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